

# Derek M. Reedy

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Full-stack cloud software engineer with a test-focused/CI/CD philosophy and a decade of enterprise software experience in AI/NLP algorithms and service infrastructure.

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| <b>Work:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Writer</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2023 to present |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Independent Sabbatical</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Cambridge, MA   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <ul style="list-style-type: none"><li>• Wrote an 80,000-word first draft of a speculative science fiction novel.</li><li>• Developed, and used AI LLM agents to extend and validate, a <math>\text{\LaTeX}</math>-based typesetting and digital publishing pipeline with a git-backed reversion system.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                 |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Lead Software Engineer</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2021 to 2023    |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Conveyor AI, powered by IBM</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Cambridge, MA   |
| <ul style="list-style-type: none"><li>• Designed, and led a team to implement, a distributed orchestration platform that enabled customers to rapidly and easily compose modular AI and business logic components into production-ready workflows.</li><li>• Implemented modern LLM applications and patterns, including agentic loops and retrieval-augmented generative QA.</li><li>• Led the product through the IBM new product launch process, meeting enterprise security, code rigor, and quality requirements.</li><li>• Engaged internal and external customers with bespoke demos and pilot projects, including a 2022 Digimarcon talk on modern AI concepts.</li></ul>                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                 |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Advisory Software Engineer</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 2015 to 2021    |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>IBM Watson Natural Language Classifier</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Cambridge, MA   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <ul style="list-style-type: none"><li>• Designed and implemented CI/CD infrastructure for kubernetes-based cloud services, coordinating the work of four internal component development teams through automation, and bringing the iteration time for releases down from a week, to days.</li><li>• Led two years of development, project management, and release on a natural language intent classification cloud service, managing new feature delivery, GDPR and HIPAA compliance, security, availability, and platform maturity through the feature's end-of-life.</li><li>• Reduced time-to-delivery of new experimental algorithmic features from months to weeks by designing and implementing a performance testing and record-keeping framework.</li><li>• Served as an SME/consultant with IBM Research teams on hardening and productizing AI research projects in natural language processing.</li></ul> |                 |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Software Engineer</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 2013 to 2015    |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>IBM Watson</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Littleton, MA   |
| <ul style="list-style-type: none"><li>• Built test automation covering components of a multi-lingual question-answering pipeline, including natural language processing algorithms, information retrieval systems, and machine learning algorithms.</li><li>• Collaborated with IBM Research and the original Jeopardy! Watson team on a feature delivery pipeline from research to product.</li><li>• Developed a noisy-channel spellchecker service, along with a testing and accuracy evaluation framework based on an automatically-generated corpus of Wikipedia edits.</li><li>• Developed tooling for visualizing test history and failure patterns, shortening daily test review to minutes.</li></ul> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                 |

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| <b>Skills:</b> | <b>Languages -</b>   | Java, Python, TypeScript (Node.js) Groovy, Elixir, LISP, Prolog, C++                                        |
|                | <b>Platform -</b>    | Kubernetes, AWS, Terraform, Docker, Hashicorp Vault, UIMA                                                   |
|                | <b>Development -</b> | Claude/Claude Code, Cucumber, Git, Jenkins, Postgres (Prismatic ORM), Redis, Kafka, Lucene/Solr, Pandas, D3 |

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| <b>School:</b>                                                                                                                                                                                                 | <b>B.S. Computer Science and Cognitive and Brain Science</b> | 2008 to 2012 |
|                                                                                                                                                                                                                | <b>Tufts University</b>                                      | Medford, MA  |
| Coursework emphasizing artificial intelligence, linguistics, logic, machine learning, natural language processing, neuroscience, philology, philosophy of language, programming languages, and theory of mind. |                                                              |              |